Assignment (12/03/2026)

Assignment Name :Decision Tree on Paper – Should I Play Outside?

Objective

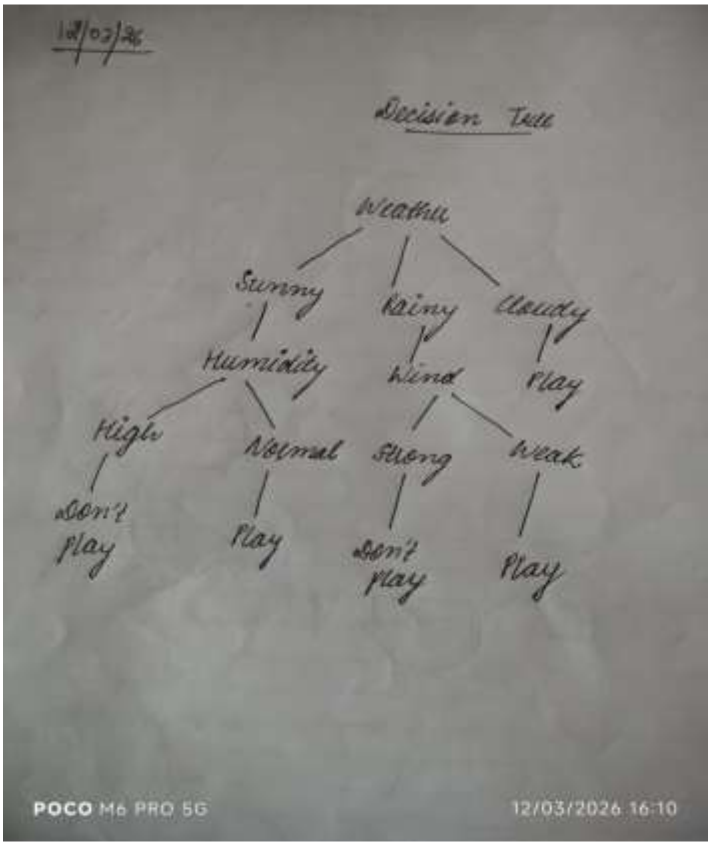
The objective of this assignment is to understand how a **Decision Tree algorithm** makes decisions by splitting data based on conditions. We will create a simple decision tree to predict whether we should **play outside or not** based on weather conditions.

Features Considered

To make the decision, we consider the following factors:

- **Weather** (Sunny / Rainy / Cloudy)
- **Temperature** (Hot / Mild / Cool)
- **Humidity** (High / Normal)
- **Wind** (Strong / Weak)

Decision Tree Structure



Explanation

1. **If the weather is Sunny**
    - If **humidity is high → Don't Play**
    - If **humidity is normal → Play**
2. **If the weather is Rainy**
    - If **wind is strong → Don't Play**
    - If **wind is weak → Play**
3. **If the weather is Cloudy**
    - **Play outside**

Conclusion

A **Decision Tree** helps make decisions step by step using conditions. Each internal node represents a **feature**, branches represent **rules**, and leaf nodes represent the **final decision**. This method is easy to understand and widely used in **classification problems in machine learning**.